

Xu Dong

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SUMMARY

- Third-year Ph.D. candidate in **Innovative Media Technology** at the **University of Surrey**.
 - Strong coding proficiency with practical experience in machine learning, deep learning, and computer vision model development.
 - 1.5 years of industry experience in applied AI, focusing on model design, optimisation, and large-scale deployment.
- Research Areas:**
- Image and video analysis
 - Action recognition and quality assessment
 - Generative AI for action and motion synthesis
 - Language model and reasoning
 - Multimodal learning

WORK EXPERIENCE

- Algorithm Engineer – Aibee Inc.** (Full-time) Mar 2022 – Aug 2023
- Developed and optimised deep learning models for intelligent retail and transportation scenarios, including event detection and multi-object tracking, with a focus on accuracy and real-time performance.
 - Built core algorithms for multi-camera vehicle tracking (1000+ vehicles/hour) to improve parking management efficiency.
 - Involved in data processing, collection, cleansing, and performance tuning across the full AI development pipeline.
- Research Assistant – University of Surrey** (Part-time) Dec 2024 – Present
- Investigating the impact of network packet loss on user experience and quality in VR-based video conferencing platforms.
 - Responsible for experiment setup, participant recruitment, data collection, and analysis.

EDUCATION

- 2023 – Present **University of Surrey, UK**
Ph.D. in Innovative Media Technology
Supervisors: Dr Andrew Gilbert, Dr Femi Adeyemi-Ejeye, and Prof Wanqing Li
- 2020 – 2021 **Queen Mary University of London, UK**
M.Sc. in Media and Arts Technology (**Distinction**)
Supervisor: Prof. Ebroul Izquierdo
- 2017 – 2020 **University of Sheffield, UK**
B.Sc. in Artificial Intelligence and Computer Science

PUBLICATIONS

- **Under Review:** X. Dong, S. Rahmaniboldaji, W. Binning, et al. "VBEST: Video Benchmark of Multi-Expertise Sporting Tasks" *Annual AAAI Conference on Artificial Intelligence (AAAI)*, 2027.
- **Under Review:** X. Dong, W. Li, A. Adeyemi-Ejeye and A. Gilbert. "BoT-Feedback: Biomechanical Reasoning for Explainable Human Action Feedback" *Annual AAAI Conference on Artificial Intelligence (AAAI)*, 2027.
- X. Dong, W. Li, A. Adeyemi-Ejeye and A. Gilbert. "Moving Beyond More Views: Redundancy-Aware Ego-Exo Fusion for Proficiency Estimation" *European Conference on Computer Vision (ECCV)*, 2026.
- X. Dong, X. Liu, W. Li, A. Adeyemi-Ejeye and A. Gilbert. "UIL-AQA: Uncertainty-aware Clip-level Interpretable Action Quality Assessment." *International Journal of Computer Vision (IJCV)*, 2025.
- X. Dong, W. Li, A. Adeyemi-Ejeye and A. Gilbert. "Generative Data Augmentation for Skeleton Action Recognition." *IEEE International Conference on Automatic Face and Gesture Recognition (FG Oral)*, 2025.

- **X. Dong**, X. Liu, W. Li, A. Adeyemi-Ejeye and A. Gilbert. "*Interpretable Long-term Action Quality Assessment.*" *British Machine Vision Conference (BMVC Oral)*, 2024.
- X. Liu, **X. Dong**, S. Qian, D. Kanojia, W. Wang, and Z. Feng. "*GCDance: Genre-Controlled Music-Driven 3D Full Body Dance Generation.*" Submitted to *IEEE Transactions on Multimedia (TMM)*, 2025.
- A. Adeyemi-Ejeye and **X. Dong**. "*Evaluating the Resilience of 2D Conferencing Platforms in VR Under Packet Loss: A Pilot Study.*" *IEEE International Conference on Consumer Electronics (ICCE)*, 2025.
- X. Kang, Y. Yuan, **X. Dong**, M. Awais, L. Tang, J. Kittler and Z. Feng. "*Short-term 3D Human Mesh Recovery with Virtual Markers Disentanglement.*" *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshop (CVPRW)*, 2025.
- W. Binning, S. Rahmaniboldaji, **X. Dong** and A. Gilbert. "*Detection and Re-Identification in the Case of Horse Racing.*" *Conference on Visual Media Production (CVMP)*, 2024.

PROJECTS

Large-scale Sports Video Understanding QA Benchmark

Oct 2025 – Present

Led a team of four to develop an LLM-automated annotation workflow to address cognitive gaps in LLMs regarding specialized sports logic, enabling the scalable generation of high-value datasets enriched with fine-grained spatio-temporal attributes and professional evaluation metrics. The system's capabilities in analyzing, diagnosing, and providing feedback for complex sports videos were significantly enhanced.

Ego-Exo Multi-View Skill Understanding System

May 2025 – Present

Developing an Ego-Exo skill understanding system that integrates redundancy-aware multiview fusion with LLM-based video question answering to enable robust assessment and interpretable feedback across heterogeneous camera views.

Music-to-Dance Generation Framework (GCDance)

Feb 2025 – Sep 2025

Developed a diffusion-based system for music-driven 3D dance generation that produces genre-consistent, realistic, and rhythm-synchronised choreography.

Generative Data Augmentation for Skeleton Action Recognition

Dec 2024 – May 2025

Developed a diffusion-based generative pipeline to synthesise high-quality 3D skeleton sequences for data augmentation, improving action recognition performance in low-data settings.

Interpretable Video-based Action Quality Assessment System

Jan 2024 – Dec 2024

Designed an interpretable video-based action quality assessment system integrating temporal modelling and visual interpretability modules to evaluate and interpret human performance.

Multi-Modal Product Identification (ACMMM 2021)

Apr 2021 – Oct 2021

Built a multi-modal product identification system using Scaled-YOLOv4, LSTM, and CGD with data augmentation and feature fusion for instance-level retrieval.

Two-Stream Figure-Ground Group Activity Recognition System

Sep 2020 – Sep 2021

Built a two-stream group activity recognition system for soccer videos using Mask-RCNN for object detection, I3D/C3D for video features, and ST-GCN for skeleton-based action recognition.

SKILLS

Programming Languages	Python, C++/C, Java, HTML, Shell.
Libraries & Frameworks	PyTorch, Tensorflow, OpenCV, NumPy, Pandas.
Tools & Platforms	Linux, Git, Docker, HuggingFace Hub, Hadoop, MySQL, Unity.
Media & Editing	Adobe Photoshop, Premiere Pro; photography and videography experience.